Hussain Abuwala

Software Developer · AI/LLM · Backend

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SUMMARY

Software developer with 3+ years of experience building AI-powered analytics products, backend services, and cloud-native applications at IBM. Progressed from Band 6 to Band 7 and delivered LLM-based systems that cut processing costs by 99.8%, improved performance 6.5x, and generated $19K in monthly customer savings.

TECHNICAL SKILLS

Languages: TypeScript, Java, Python, Bash

AI: LLMs, Prompt engineering, Agents, Embeddings

Backend: Node.js, RabbitMQ, SQLite, nginx

Frontend: React

Infrastructure: Docker, Kubernetes, Linux, Git, k6

EXPERIENCE

Software Developer, Band 7, IBM, Toronto, Canada — Oct 2025 – Present

- Rebuilt IBM watsonx BI metadata enrichment (v2) as a native LLM pipeline, cutting per-asset cost 99.8% ($5.32 to $0.009), delivering $19K in monthly customer savings, maintaining 90%+ accuracy, and improving performance 6.5x.
- Developed Data Stories, an autonomous analytics agent that surfaced 2-3 decision-grade, verifiable findings from a dataset without requiring users to formulate questions; selected for executive review.
- Implemented Custom Instructions, enabling business users to encode domain rules and knowledge in natural language; integrated the instructions into asset discovery and SQL generation to produce answers grounded in each customer's business context.
- Built a React visualization editor for IBM watsonx.data with manual authoring and rule-based chart recommendations in the browser, inferring measures and categories from column types, values and SQL grouping, at no model cost.

Software Developer, Band 6, IBM, Toronto, Canada — Jan 2024 – Sep 2025

- Modernized a legacy metadata enrichment service (v1) and integrated watsonx.data intelligence classification, reducing glossary-artifact upload time from approximately 20 minutes to under 3 minutes; powered five core services spanning conversational analytics, visualization, insights, semantic search, and Suggested Questions.
- Built an LLM-powered Suggested Questions service that used enriched schemas to generate and pre-validate answerable analytical questions, creating a guided first interaction for users; recognized with an IBM Outstanding Innovation Award.

PROJECTS

URL Shortener | TypeScript, Node.js, SQLite, nginx, Litestream, Linux, Bash — Oct 2026

- Built a URL shortener on a single server and load-tested it with k6 through nine scripted events up to 10% of Bitly's traffic (1,000 redirects/s); final design serves an estimated 877M redirects/month at 14.8 ms p99 with 0% errors for EUR 9.56/month, and recovers from machine loss in under 6 minutes. hussainabuwala.github.io/projects/url-shortener/

LLM from scratch | Python, NumPy, Jupyter, Matplotlib, unittest — Oct 2026

- Building a character-level language model from scratch in a 14-episode video series (Python, NumPy, later PyTorch), from count-based n-grams to a decoder-only Transformer; each episode ships tested code and notebooks. hussainabuwala.github.io/projects/llm-from-scratch/

EDUCATION

University of Waterloo, Master of Mathematics (MMath) in Computer Science; CGPA 94/100 — Jan 2022 – Apr 2023

Monash University, Bachelor of Computer Science (Honours); CGPA 4.0/4.0; Overall Best Graduate (100+ cohort); High Achiever Award (top 1%) — Feb 2016 – Jun 2020